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Are Antidepressants Just Treating Protein Deficiency?

This one amino acid outperforms antidepressant drugs in clinical trials but protein itself would be even better.



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Getting protein up to 120 grams per day for women and 160 grams per day for men would likely massively outperform antidepressant drugs just by itself, and these targets form a central foundation of the [Depression Protocol](#).

While this relates to much more than tryptophan, getting enough tryptophan is certainly part of it, so we start by looking at the randomized controlled trials of this one powerful little amino acid.

In [the first](#) of two tryptophan trials, 115 subjects were randomly allocated to one of four treatments: three grams of tryptophan per day (in three divided doses of one gram each), 150 milligrams of the tricyclic antidepressant amitriptyline (in three divided doses of 50 milligrams each), a combination of both of those treatments, or to placebo.

All treatments outperformed the placebo group, with tryptophan-only performing identically to amitriptyline-only, and their combination modestly outperforming each monotherapy.

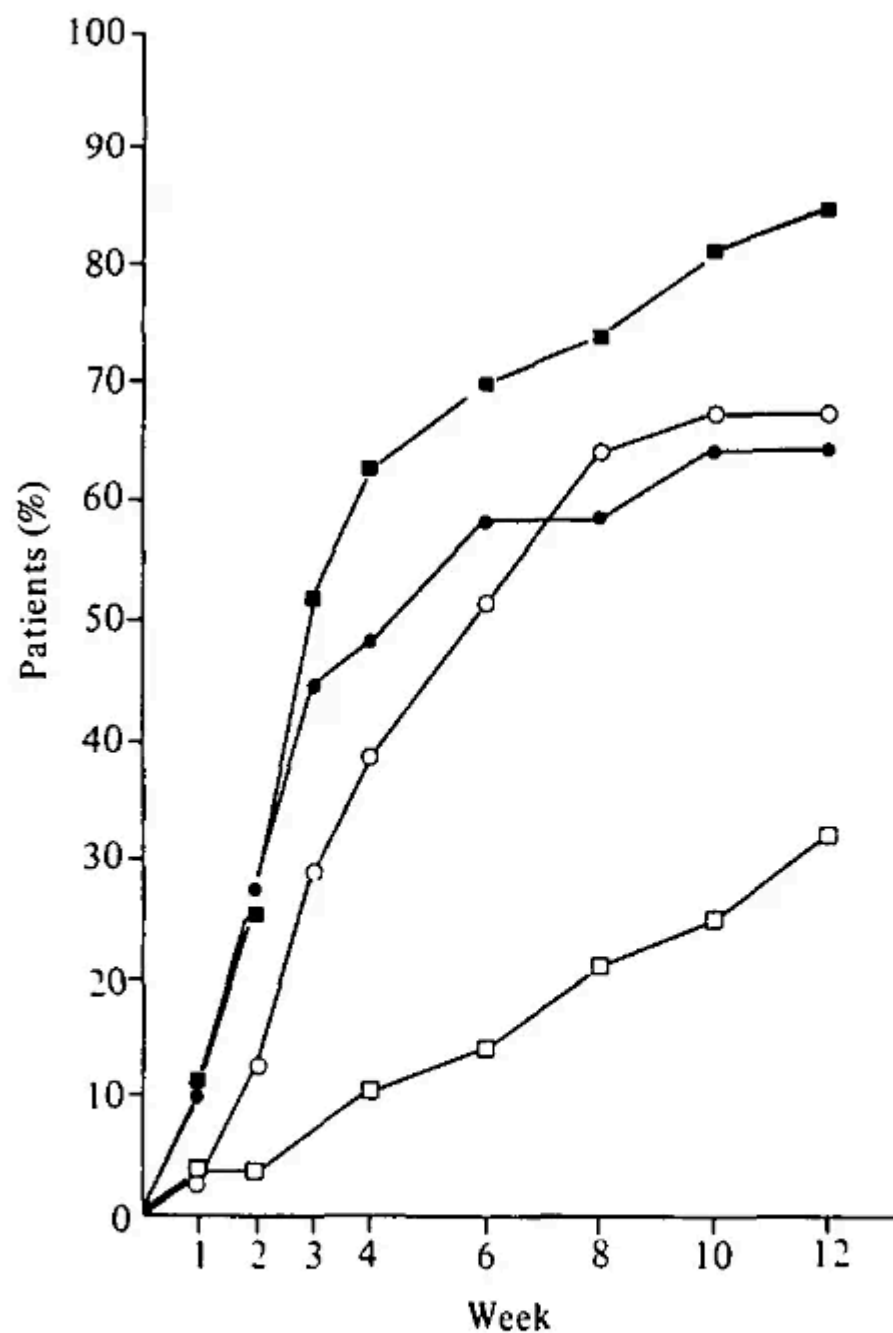


FIG. 1. Comparison of treatments showing the cumulative % of patients in each group in remission, defined as a fall to 4 points or less on the total HDS score. ●—●, tryptophan; ○—○, amitriptyline; ■—■, tryptophan amitriptyline combination; □—□, placebo.

The number of subjects withdrawing due to side effects was twice as high in the amitriptyline-only group (7) than in the tryptophan-only group (3), but was zero in the placebo group and was otherwise lowest in the combination therapy group (1). This gives the impression that amitriptyline is considerably more dangerous than tryptophan when used alone but together they each protect against each other's respective side effects.

Since tryptophan was just as effective as amitriptyline but had less than half the rate of serious side effects, tryptophan outperformed amitriptyline from a cost-benefit standpoint.

I propose the following for why they both protect against each other's side effects.

Amitriptyline inhibits serotonin transport like SSRIs do, so probably has largely similar effects on serotonin. It could be depleting circulating serotonin and whole-body serotonin, and extra tryptophan could be making up for this by allowing re-synthesis of intracellular serotonin. By diverting tryptophan into serotonin synthesis, moreover, it is inducing a deficiency of tryptophan itself. Since tryptophan is the precursor to niacin, NAD⁺, and NADPH, amitriptyline could be hurting mitochondrial energy metabolism and antioxidant defense by inducing a deficiency of tryptophan and its derivatives.

Meanwhile, excess tryptophan without the amitriptyline could be metabolized into quinolinic acid, an excitatory neurotransmitter that carries the risk of excessive neurostimulation or excitotoxicity, and to glutaryl CoA, an intermediate in the catabolism of tryptophan that is toxic to mitochondrial energy metabolism and creates an inflammatory stimulus to help with its detoxification.

By increasing the demand for tryptophan, amitriptyline effectively lowers the dose of tryptophan below that which would cause accumulation of quinolinate and glutaryl CoA, and by increasing the supply of tryptophan, tryptophan protects against amitriptyline-induced deficiencies of whole-body and intracellular serotonin and of niacin and its derivatives.

If this explanation is correct, it suggests that **3 grams of tryptophan is an overdose, and that a lower dose such as 1-2 grams per day might be just as effective and safe as the tryptophan/amitriptyline combination.**

The remission rate was 81% in the combination group. Thus, over 80% of depression might disappear with simply optimizing tryptophan status!

In [the second tryptophan trial](#), 30 subjects were randomly allocated to fluoxetine (Prozac) with tryptophan or a placebo, but no one was randomized to a placebo-only or tryptophan-only group. The tryptophan was titrated up from one gram to four grams over the first four weeks. Everyone received 20 milligrams of Prozac daily. The trial lasted eight weeks. The two treatments performed identically, decreasing average depression scores by about 40%. Five people dropped out of the Prozac plus tryptophan group and four dropped out of the Prozac-only group, but most of these were unrelated to treatment. The one exception was that one person in the Prozac-only group developed new-onset psychosis.

The second tryptophan trial is not very relevant to our analysis as it didn't examine the impact of tryptophan without a drug, and if the first trial overdosed the tryptophan the second trial overdosed it even more. The most robust point here is drawn from the first trial, which is that tryptophan alone works just as good as antidepressant drugs and is safer.

Tryptophan is the precursor to 5-HTP, which is the precursor to serotonin, so 5-HTP trials are also relevant.

A [small trial of 5-HTP](#) from 1972 with only ten people in it found that three grams of 5-HTP per day for three weeks increased the probability of improvement [13-fold](#) as calculated in a meta-analysis, but what the raw data show is that three people improved in the 5-HTP group and none in the placebo group. Several earlier studies reviewed in this paper showed no effect of much smaller amounts of intravenous 5-HTP.

More recently, [60 patients](#) were randomly allocated to eight weeks of 5-HTP or fluoxetine (Prozac). The 5-HTP was titrated up from 150 milligrams per day to 400 milligrams per day in three divided doses over the first four weeks, while the fluoxetine was titrated up from 20-40 milligrams per day over the same period. Both

treatments decreased average depression scores by close to 70% and were statistically indistinguishable from one another.

A similar trial of [69 patients](#) showed 100 milligrams of 5-HTP performed identically to fluvoxamine over six weeks, with average depression scores getting cut more than in half.

While 5-HTP may seem “similar” to SSRIs in that it is a precursor to serotonin, it is grossly misunderstanding SSRIs to consider them similar.

First, SSRIs massively deplete circulating serotonin and most likely massively deplete whole-body serotonin outside the brain and gut. SSRIs that have poor activation of the sigma-1 receptor even deplete brain serotonin.

Second, SSRIs block the entry of serotonin into the cell, which is absolutely critical to its function, whereas 5-HTP gives the cell what it needs to produce serotonin within itself.

Third, a precursor does not *force* serotonin to be at the synapse, whereas an SSRI does. Thus, the precursor will not have anywhere near the risk of dependence on the unnaturally high concentration or the unnaturally long duration of synaptic serotonin that a drug can cause.

Drugs do not normalize anything. They make things abnormal in ways that nutrients — which are just as effective or more effective — cannot do.

Nevertheless, it is much more natural to obtain tryptophan from whole-food protein than to take 5-HTP.

The tryptophan and 5-HPT trials did not examine the dietary intake of protein, which is a pretty unforgivable oversight for trials examining the impact of supplementing with an amino acid or its derivative.

There is some variation in the tryptophan content of food proteins, but as long as you consume enough total protein and don't count tryptophan-poor collagen toward it, you will get enough. The average [tryptophan intake](#) in the US is 826 milligrams per day, 977 milligrams for men and 679 milligrams for women. However, the bottom 10% only get 502 and the top 10% get 1410 milligrams, which is almost three times higher. Self-rated depression goes down with increasing tryptophan intake, but most of that association is found at lower-than-average tryptophan intakes, indicating that many depressed people will need an extra 300 milligrams of tryptophan per day.

Total protein intake and protein quality are far more sensible to use as a primary analysis because neurotransmitters other than serotonin are important to depression but derived from other amino acids. For example, dopamine and norepinephrine are derived from phenylalanine and tyrosine. The ability to methylate dopamine is supported by methionine.

Observationally, each 10% increase in [calories from total protein](#) is associated with a close to 40% lower risk of depression, up through at least 25% of calories.

Using average reported calories of 1900 per day for women and 2500 for men, the 25% figure works out to 118 grams per day for women and 156 grams per day for men.

Percent of calories is a ridiculous and biologically irrelevant way to analyze protein, which likely made the correlation weaker than it would have been if adjusted for lean body mass or weight. Therefore, **the average woman should consume at least 118 grams of non-collagen protein and the average man should consume at least 156 grams of non-collagen protein per day** to minimize the risk of depression.

These should be adjusted by body mass based on 2.2 grams of protein per kilogram bodyweight on the basis of the classical “standard reference man” representing an adult male who hasn’t been as subject to the obesity epidemic as is the average modern man.

Thus, the 118/156 figures or their rounded counterparts of 120 and 160 should be used as a rule of thumb, but if your bodyweight is substantially different than average due to your height or lean mass or if you are adjusting for children, use 2.2 grams of protein per kilogram of ideal body mass. Use lean mass if you have an unusually high lean mass for your height and you have an estimate of it from a DEXA.

Collagen is important to consume, but is unique in its low quantity of tryptophan, so collagen should be consumed over and on top of this minimum intake of non-collagen protein just described.

Some people report these high protein intakes causing insomnia or some other similar problem. If high protein intake causes problems with insomnia, anxiety, mania, or anything else that looks like overstimulation, the appropriate thing to do is use the [Sulfur Protocol](#), but the [Depression Protocol](#) does the heavy lifting by extracting the most useful strategies from the Sulfur Protocol to deal with these issues.

It also covers contingencies for methionine/glycine balance and glutaryl CoA accumulation.

Tryptophan is the precursor to 5-HTP, and comparing the trials above would imply they are equally effective. 5-HTP is converted to serotonin, and serotonin is converted to melatonin, which is not only needed in circulation to promote healthy sleep but is needed inside mitochondria to act as an antioxidant and to support energy metabolism during relative hypoxia.

The ability to support 5-HTP synthesis from tryptophan is complicated by a large number of nutritional cofactors, and while it does not occur in the mitochondria, it does depend on mitochondrial energy metabolism.

The ability to convert 5-HTP into serotonin is far more simple and only requires vitamin B6.

The conversion of serotonin to melatonin is dependent on vitamin B5 (pantothenic acid) in the form of coenzyme A and methylation.

So, it is possible that someone could get enough tryptophan from dietary protein but could still have their ability to make 5-HTP limited by any number of nutrient deficiencies. It is best to ensure adequate total protein intake and resolve any

micronutrient deficiencies using the [Comprehensive Nutritional Screening](#) and any deficits in mitochondrial ATP production using your personalized [Mitome](#) protocol prior to resorting to 5-HTP, but 5-HTP could be tried if there is a suspected genetic problem with its synthesis or if it proves beneficial even after resolving all of these issues.

However, the trials above support equivalency between tryptophan and 5-HTP for the average depressed person, indicating that the average depressed person simply needs more tryptophan instead of 5-HTP, and most of them could probably get the tryptophan they need by eating more protein.

Eating more protein would be the ideal solution because it would support the other amino acid-based neurotransmitters and the methylation system alongside optimizing tryptophan status.

120 grams of protein is 480 Calories, so the average woman who consumes 1900 Calories has plenty of room for this.

160 grams of protein is 640 Calories, so the average man who consumes 2500 Calories has plenty of room for this.

The way to eat more non-collagen protein is to eat more animal flesh, dairy, eggs, and legumes. Even on a vegan diet you can hit these targets with lentils or tofu, and it is all the easier on an omnivorous or carnivorous diet.

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♥ Liked by Chris Masterjohn, PhD

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Thank you Chris for publishing information about this connection. I'm living proof that what you say is indeed correct. Depression in my 20s. Terrible depression in my 30s after going vegetarian. Resolved completely after going strict paleo in my 50s when significantly increasing my protein intake. I woke up one morning a few months in and realized my depression was gone. I attributed it to more protein and saturated fat.

I've sung this from the rooftops but no one believes me. 🙄 now I can point them to your article.

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My child's schizoaffective disorder was fully reversed by going high protein, low carb. Paleo is the maintenance plan. Three years symptom free and on zero meds.

Like Lawandorder said, no one believes me or wants to hear about it. I am grateful for our fabulous doctor who saw another way other than meds with their horrible side effects.

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